



Lecture 20

Let's solve some example problem.

For the given transfer function, specify the gain k so that the output $y(t)$ has an overshoot of no more than 10% in response to a unit step.

$$\frac{Y(s)}{R(s)} = \frac{k}{s^2 + 2s + k} = G(s)$$

Let's compare $G(s)$ with the standard transfer function for a second

order system.

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n = \sqrt{k}$$

$$2\zeta\omega_n = 2$$

$$\zeta = \frac{1}{\omega_n} = \frac{1}{\sqrt{k}}$$

$$M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \leq 0.1$$

$$\ln M_p = -\frac{\zeta\pi}{\sqrt{1-\zeta^2}} = 0.1$$

$$\Rightarrow (\ln M_p)^2 = \frac{\zeta^2 \pi^2}{1-\zeta^2}$$

$$\Rightarrow (\ln M_p)^{\sim} - (\ln M_p)^{\sim} \xi^{\sim} = \xi^{\sim} \pi^{\sim}$$

$$\Rightarrow (\ln M_p)^{\sim} = \xi^{\sim} (\pi^{\sim} + (\ln M_p)^{\sim})$$

$$\Rightarrow \xi = \sqrt{\frac{(\ln M_p)^{\sim}}{\pi^{\sim} + (\ln M_p)^{\sim}}}$$

$$\Rightarrow \xi = 0.591$$

As a result, not to exceed 10% overshoot, $\xi \geq 0.591$

$$k = \frac{1}{\xi^{\sim}} = 2.86$$

Let's say the following transfer function is given:

$$\frac{Y(s)}{R(s)} = \frac{100k}{s^{\sim} + (25+a)s + 25a + 100k}$$

Find the value of a and K
so the following specification is
met

$$M_p \leq 25\%$$

$$t_s \leq 0.1 \text{ sec}$$

Let's find the value of ζ first.

$$M_p = e^{-\zeta \pi / \sqrt{1-\zeta^2}}$$

$$\zeta = 0.4037$$

for overshoot not exceed 25%,

$$\zeta \geq 0.4037$$

We know that

$$e^{-\zeta \omega_n t_s} = 0.01, \quad t_s = 0.1$$

$$\zeta = 0.4031$$

using the above equation

$$\omega_n = 44.07$$

Now we compare with the standard transfer function.

$$\frac{Y(s)}{R(s)} = \frac{100k}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$2\zeta\omega_n = 25 + a$$

$$\begin{aligned}\Rightarrow a &= 2\zeta\omega_n - 25 \\ &= 92.1 - 25 \\ &= 67.1\end{aligned}$$

$$\omega_n^2 = 25a + 100k$$

$$k = \frac{\omega_n^2 - 25a}{100} = 443.34$$

